



Indian Institute of Technology Hyderabad

Assignment 4 (Lab Report)

Basic BioInformatic (BM2023)

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Basic Bioinformatics Lab Report-4

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Part 1

A. The properties of a good primer pair for real time PCR.

Primer Length

Primers typically range between 18 and 25 nucleotides. This length ensures specificity while maintaining efficient annealing.

- **Melting Temperature (T_m)**

The melting temperature should generally lie between 55°C and 65°C. Forward and reverse primers should have similar T_m values within a small range to ensure uniform binding.

- **GC Content**

The GC content should ideally lie between 40% and 60%. This provides stable hybridization while avoiding excessively strong binding.

- **Absence of Secondary Structures**

Primers should not form hairpins, self dimers, or cross dimers, as these structures interfere with amplification efficiency.

- **Amplicon Length**

For real time PCR experiments, the optimal amplicon size typically ranges between 100 and 200 base pairs.

- **Target Specificity**

Primers should uniquely bind to the desired target region in the genome to avoid non specific amplification.

Find a primer pair by using primer-BLAST for GLUT-4 of human

FASTA of GLUT-4 of Human

MPSGFQQIGSEEDGEPPQQRVTGTLVLAVFSAVLGSLQFGYNIGVINAPQKVIEQ
SYNETSIPPGLTTLVVALSVAIFSVGGMISSFLIGHISQWLGRKRAMLVNVL
VLGGSLMGLANAASYEMLILGRFLIGAYSGLTSGLVPMYVGEIAPTHLRGALGT
LNQLAIVIGILIAQVLGLESLLGTASLWPLLLGLTVLPALLQLVLLPFCPESPR
YLYIIQNLEGPARKSLKRLTGWADVSGVLAELKDEKRKLERERPLSLLQLLGS
RTHRQPLIIAVVLQLSQQLSGINAVFYYSTSIFETAGVGGPAYATIGAGVVNTVF
TLVSVLLVERAGRRTLHLLGLAGMCGCAILMTVALLLLERVPAMSYVSIVAIFG
FVAFFEIGPGPIPWFIWAELFSQGPRPAAMAVAGFSNWTSNFIIGMGFYVAEA
MGPIYVFLLEAVLLLGGFFITFLRVPETRGRTFDQISAAFHRTPSLLEQEVKPS
T
ELEYLGPDEND

Many Primers were found during the Primer BLAST

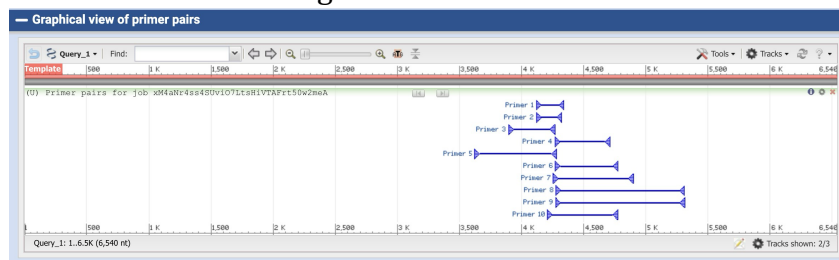


Figure 1: Primer BLAST

Top 5 Primer Pairs are attached below :

Primer pair 1									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TCTCCAACGGACGAGCAAC	Plus	20	4139	4158	59.97	55.00	6.00	0.00
Reverse primer	GGTCCACCAAACTGACCCAA	Minus	20	4320	4301	60.11	55.00	4.00	0.00
Product length	182								
Primer pair 2									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CTCCAACGGACGAGCAACT	Plus	20	4140	4159	59.97	55.00	6.00	1.00
Reverse primer	AAAGACCATGCTGTGGC	Minus	20	4302	4283	60.25	55.00	8.00	2.00
Product length	163								
Primer pair 3									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CTCTGAGTTGAGGCAAGGG	Plus	20	3916	3935	60.04	60.00	5.00	0.00
Reverse primer	CTGTGTGATGCCACCTCTG	Minus	20	4247	4228	60.68	60.00	4.00	1.00
Product length	332								
Primer pair 4									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	AGACCATGGGTCTTTGGGTC	Plus	20	4288	4307	59.30	55.00	8.00	2.00
Reverse primer	GAGGTGTTGACGGAGTGAGG	Minus	20	4690	4671	60.04	60.00	2.00	0.00
Product length	403								
Primer pair 5									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CTCCTGAGGGATTGGGCTG	Plus	20	3643	3662	60.11	60.00	5.00	1.00
Reverse primer	AGGTGGGCTAGCTGTGTGAT	Minus	20	4258	4239	60.91	55.00	6.00	2.00
Product length	616								

Figure 2: Top 5 Primer BLAST result

B. Optimal primer from primer blast

Primer pair 1									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TCTCCAACGGACGAGCAAC	Plus	20	4139	4158	59.97	55.00	6.00	0.00
Reverse primer	GGTCCACCAAACTGACCCAA	Minus	20	4320	4301	60.11	55.00	4.00	0.00
Product length	182								

Figure 3: Selected Primer for further analysis

C. BLAST Analysis Against the Whole Genome

Organism	Blast Name	Score	Number of Hits	Description
root			102	
• artificial sequences	other sequences		9	
• Human ORFeome Gateway entry vector	other sequences	40.1	1	Human ORFeome Gateway entr
• synthetic construct	other sequences	40.1	6	synthetic construct hits
• eukaryotic synthetic construct	other sequences	40.1	2	eukaryotic synthetic construct hit
• Saimiri boliviensis boliviensis	primates	40.1	1	Saimiri boliviensis boliviensis hit
• Homo sapiens	primates	40.1	13	Homo sapiens hits
• Cercopithecus atys	primates	40.1	1	Cercopithecus atys hits
• Aotus nancymaae	primates	40.1	1	Aotus nancymaae hits
• Sapajus apella	primates	40.1	2	Sapajus apella hits
• Erinaceus europaeus	insectivores	40.1	3	Erinaceus europaeus hits
• Pongo pygmaeus	primates	40.1	1	Pongo pygmaeus hits
• Cebus imitator	primates	40.1	2	Cebus imitator hits
• Mandrillus leucophaeus	primates	40.1	1	Mandrillus leucophaeus hits
• Papio anubis	primates	40.1	1	Papio anubis hits
• Pongo abelii	primates	40.1	2	Pongo abelii hits
• Macaca thibetana thibetana	primates	40.1	1	Macaca thibetana thibetana hits
• Mormoops megalophylla	bats	40.1	1	Mormoops megalophylla hits
• Theropithecus gelada	primates	40.1	1	Theropithecus gelada hits
• Macaca nemestrina	primates	40.1	1	Macaca nemestrina hits
• Macaca mulatta	primates	40.1	1	Macaca mulatta hits
• Macaca fascicularis	primates	40.1	1	Macaca fascicularis hits
• Chlorocebus sabaeus	primates	40.1	1	Chlorocebus sabaeus hits
• Gorilla gorilla gorilla	primates	40.1	1	Gorilla gorilla gorilla hits
• Callithrix jacchus	primates	40.1	2	Callithrix jacchus hits
• Mus musculus musculus	rodents	40.1	2	Mus musculus musculus hits
• Streptococcus suis	firmicutes	38.2	46	Streptococcus suis hits
• Show organism report for 2 Sapajus apella hit(s)	a-proteobacteria	38.2	2	Vicinamibacteria bacterium hits
• Mesorhizobium sp. INR15	a-proteobacteria	38.2	1	Mesorhizobium sp. INR15 hits

Figure 4:

Part 2

Find the protein details in RCSB homepage, a part of PDB site. Send the 4 letter protein code and send few details of the protein. Four Letter Protein Code : 7WSM

 **7WSM** | **pdb_00007wsm** ⓘ

Cryo-EM structure of human glucose transporter GLUT4 bound to cytochalasin B in lipid nanodiscs

PDB DOI: <https://doi.org/10.2210/pdb7WSM/pdb> **EM Map** EMD-32760: [EMDB EMDDataResource](#)

Classification: **TRANSPORT PROTEIN**

Organism(s): Homo sapiens

Expression System: Homo sapiens

Mutation(s): No ⓘ

Membrane Protein: Yes ⓘ    

 **7WSM** | **pdb_00007wsm** ⓘ

Cryo-EM structure of human glucose transporter GLUT4 bound to cytochalasin B in lipid nanodiscs



Protein Type: Integral membrane protein, part of the GLUT family (SLC2A4 gene).

Location: Found in fat (adipose) and muscle cells (skeletal & cardiac muscle).

Function: Regulates glucose uptake in response to insulin signalling.

Structure: Composed of 12 transmembrane helices, which form a channel-like pathway for glucose transport.